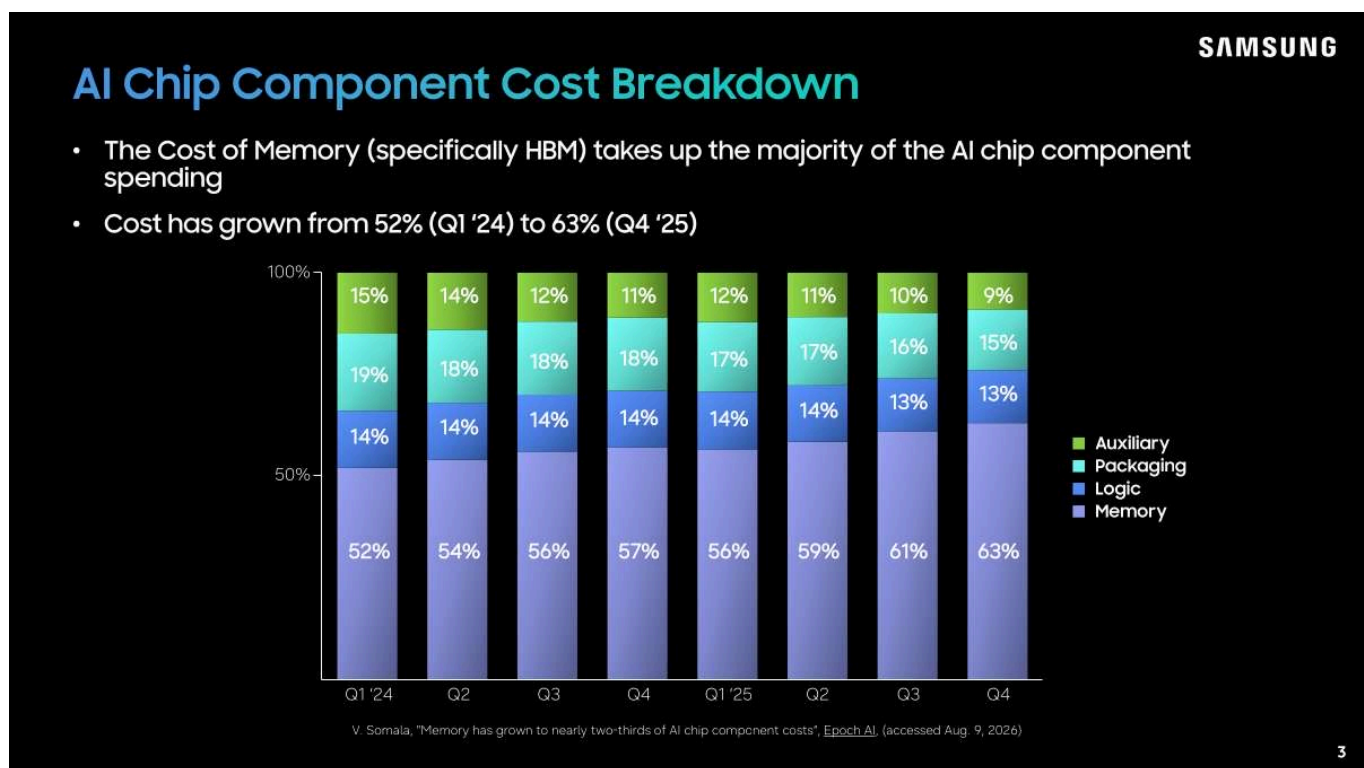


Hot Chips 2026 Samsung LPDDR5X PIM Slide 6 LPDDR5X-PIM Architecture

Samsung took the stage at Hot Chips 2026 to present LPDDR5X-PIM, a processing-in-memory memory solution built on LPDDR5X DRAM for AI inference. This is one that several folks outside the theatre were hotly anticipating today.

We are covering this talk live, so please excuse any typos.

Samsung opens with a market-context figure that frames its pitch around memory cost as a percentage of AI packages. HBM now claims the majority of AI chip component spending, and that share grew from 52 percent in Q1 2024 to 63 percent by Q4 2025. It is likely higher now.



Hot Chips 2026 Samsung LPDDR5X PIM Slide 3 AI Chip Component Cost Breakdown

HBM delivers the bandwidth AI inference needs, but with high power consumption and complex packaging. Samsung argues the route to lower cost runs through a new memory form factor the LP-PIM.

The Need for New Memory Solutions

The de facto memory solution to support AI Inference: HBM

- High power consumption
- Complex packaging



To achieve lower cost, various advancements have been made in AI Inference:

- Efficient LLM architecture
- Smaller size models



A new memory form factor is required → **LP-PIM**

- Higher bandwidth (compared to normal LPDDR)
- Lower overall power consumption at system level

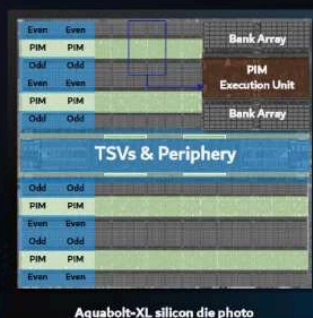
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Hot Chips 2026 Samsung LPDDR5X PIM Slide 4 The Need for New Memory Solutions

Here Samsung traces its PIM timeline, from the Aquabolt-XL HBM2-PIM proof of concept in 2021 through LPDDR5X-PIM productization in 2026 as the world's first LPDDR-based PIM solution. You can read our pieces on [Samsung HBM2-PIM and Aquabolt-XL at Hot Chips 33](#) and [Samsung Processing in Memory Technology at Hot Chips 2023](#).

Progression of Samsung PIM Technology

2021



Aquabolt-XL silicon die photo

HBM-PIM Proof of Concept

Aquabolt-XL : Samsung HBM2-PIM with in-memory processing for ML accelerators and beyond

2023

HBM PIM-Cluster



HBM-PIM Cluster Validation & LP-PIM Conceptualization

Samsung PIM/PNM for Transformer based AI

2026



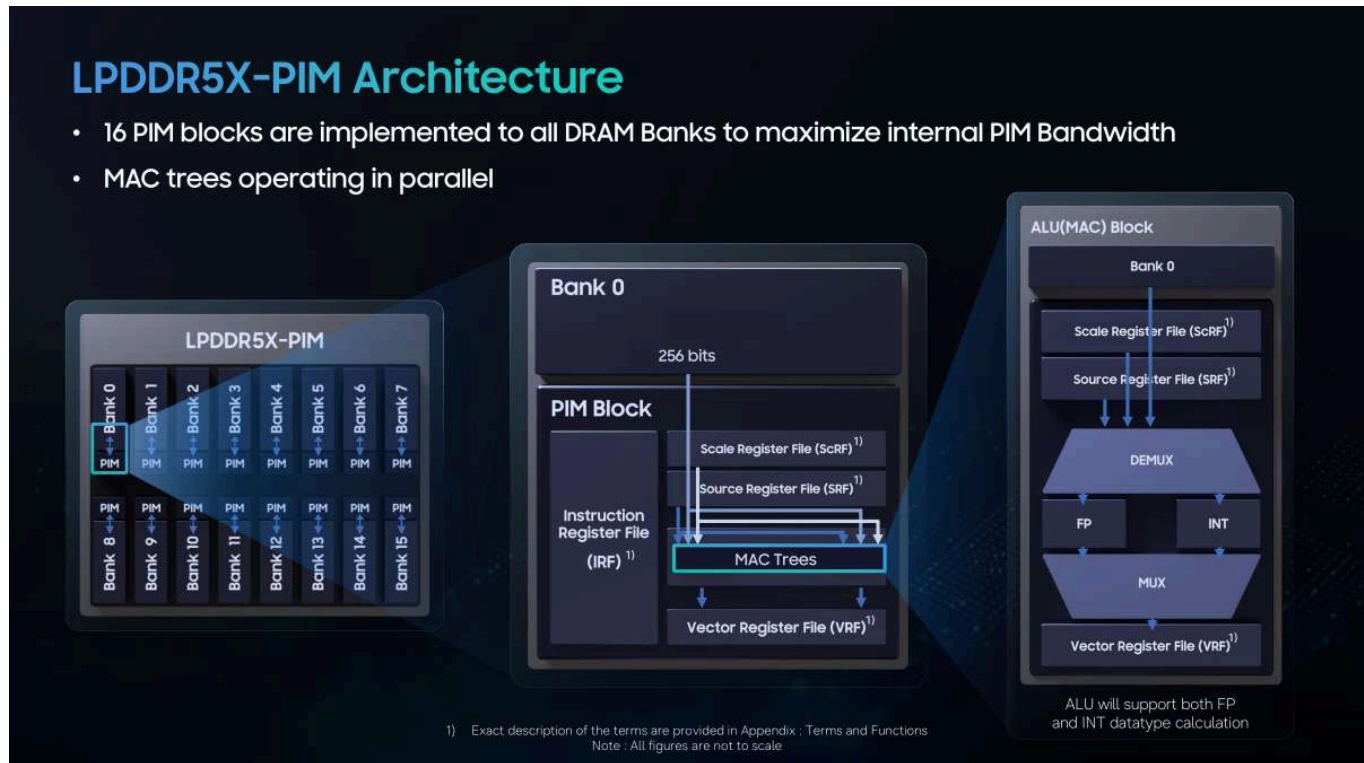
LPDDR5X-PIM Productization

World's First LPDDR-PIM Solution for AI Inference

5

Hot Chips 2026 Samsung LPDDR5X PIM Slide 5 Progression of Samsung PIM Technology

Now to the LPDDR5X-PIM architecture itself. Sixteen PIM blocks sit in the DRAM banks, with MAC trees running in parallel and an ALU that handles both FP and INT data-type calculations.



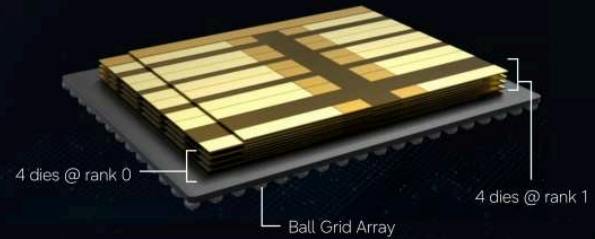
Hot Chips 2026 Samsung LPDDR5X PIM Slide 6 LPDDR5X-PIM Architecture

This specification figure shows a JEDEC-standard 561-ball package with 16 GB across four dies per rank, targeting server, mobile, and client. PIM bandwidth reaches 614 GB/s at the x64 9600 Mbps operating point, eight times the 76.8 GB/s available on the conventional DRAM side.

LPDDR5X-PIM Specifications

LPDDR5X-PIM Package

561-ball Package (based on JEDEC standard)
16 GB Density
2 ranks of 4-CH (x64)



Target Application	Server / Mobile / Client
PIM configuration	1 PIM / 1 Bank
DRAM / PIM bandwidth	76.8 GB/s ¹⁾ , 614 GB/s (8x)
Configuration Register ²⁾	256 bit / 1 PIM
Instruction Register File (IRF) ²⁾	1 kbit / 1 PIM
Source Register File (SRF) ²⁾	4 kbit / 1 PIM
Scale Register File (ScRF) ²⁾	2 kbit / 1 PIM
Vector Register File (VRF) ²⁾	1 kbit / 1 PIM

¹⁾ Based on a x64 LP5X/LP5X-PIM package operating @ 9600 Mbps frequency
²⁾ Exact description of the terms are provided in Appendix : Terms and Functions

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Hot Chips 2026 Samsung LPDDR5X PIM Slide 7 LPDDR5X-PIM Specifications

LPDDR5X-PIM is the first LP-PIM product with multi-precision data type support. Fifteen combinations are selectable through the MAC precision fields in the configuration register. Activations can pair with SINT4 weights for 2.4 TOPS, or fall back to roughly 1.2 TFLOPs per package in FP8.

LPDDR5X-PIM Multi-Precision Datatype Support

- **The first LP-PIM product with multi-precision datatype support**
- Can be selected via the 'MAC Precision Datatype Select' and 'MAC Precision Activation Bitwidth' field in the CRF
 - 15 different combinations are supported

Input Data Precision (Activation)	Input Data Precision (Weight)	Computation Power (per PKG)
UINT (8, 16)	SINT4	2.4 TOPs
SINT (4, 8, 16)		
UINT (8, 16)	SINT8	1.2 TOPs
SINT (8, 16)		
FP16	FP8 (E4M3)	1.2 TFLOPs
FP8 (E4M3, E5M2)		
FP16	FP8 (E5M2)	1.2 TFLOPs
FP8 (E4M3, E5M2)		

8

Drop-in compatibility rests on Address Align Mode. This mapping between DRAM addresses and MAC instructions lets LPDDR5X-PIM work with a conventional DRAM controller rather than requiring a new one.

SAMSUNG

How PIM Utilizes Conventional DRAM Controller

- Utilization of conventional DRAM controller is achieved through Address Align Mode (AAM)
- This novel method leverages the mapping between DRAM addresses and MAC instructions, maintaining re-ordering technique in DRAM controllers to maximize performance

Conventional (non-AAM) Instruction Case

	DRAM command	Instruction Register File	
	ACT BANK 00, RA 00		
①	READ BANK 00, CA 00	MAC VRF[0] += CA [00] * SRF[0]	ALIGN
②	READ BANK 00, CA 01	MAC VRF[0] += CA [01] * SRF[1]	ALIGN
③	READ BANK 00, CA 03	MAC VRF[0] += CA [03] * SRF[2]	MIS- ALIGN

BANK

① ② ③

0 1 2 3 CA

① ② ③ SRF

MAC

PIM BLOCK

0 1 2 VRF

AAM Instruction Case

	DRAM command	AAM Instruction Register File		
		IRF register will be first set as "Don't Care"	VRF/SRF address set based on RA/CA address, respectively	
	ACT BANK 00, RA 00			
①	READ BANK 00, CA 00	MAC VRF[*] += CA [00] * SRF[*]	MAC VRF[0] += CA [00] * SRF[00]	ALIGN
②	READ BANK 00, CA 01	MAC VRF[*] += CA [01] * SRF[*]	MAC VRF[0] += CA [01] * SRF[01]	ALIGN
③	READ BANK 00, CA 03	MAC VRF[*] += CA [03] * SRF[*]	MAC VRF[0] += CA [03] * SRF[03]	ALIGN

BANK

① ② ③

0 1 2 3 CA

① ② ③ SRF

MAC

PIM BLOCK

0 1 2 VRF

S. Lee et al., "Hardware Architecture and Software Stack for PIM Based on Commercial DRAM Technology", 2021 ACM/IEEE 48th Annual International Symposium on Computer Architecture (ISCA), 2021, pp. 43-56. Available on the IEEE Xplore Digital Library.

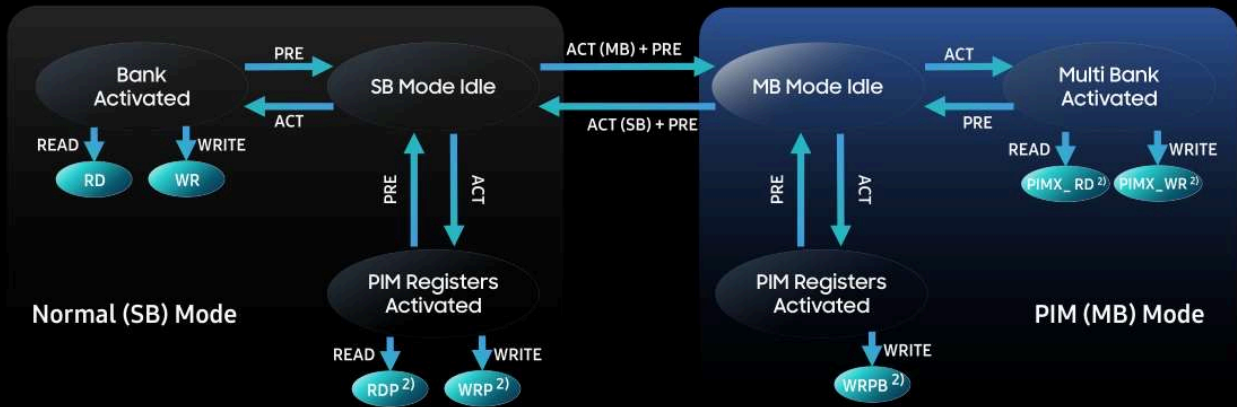
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Hot Chips 2026 Samsung LPDDR5X PIM Slide 9 How PIM Utilizes Conventional DRAM Controller

Mode change shuttles the part between single-bank and multi-bank PIM operation. Predefined rows and PIM registers handle the switch, and Samsung says this method is faster and more reliable than the HBM-PIM approach of prior generations.

LPDDR5X-PIM Mode Change

- Requires predefined rows for mode change and PIM register (Defined through MR ¹⁾)
 - Predefined rows for mode change per channel (MB: Multi-bank, SB: Single-bank)
 - A predefined row for register access per bank (PIM Registers)
 - A faster and reliable mode change method compared to previous generation (HBM-PIM)



1) MR : Mode Register
2) Exact description of the commands are provided in Appendix : Terms and Functions

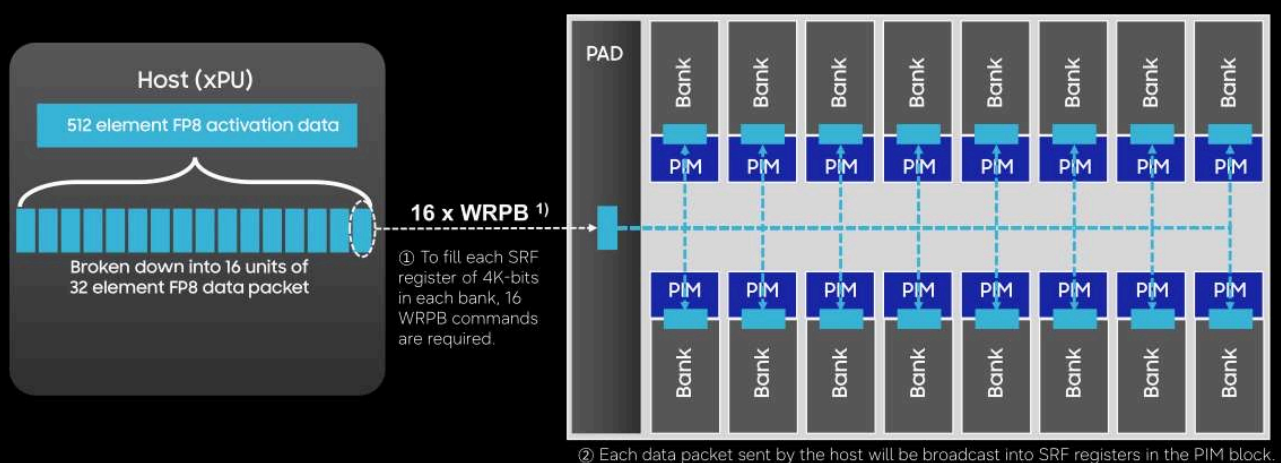
10

Hot Chips 2026 Samsung LPDDR5X PIM Slide 10 LPDDR5X-PIM Mode Change

Activation write begins the MAC flow. Sixteen WRPB commands fill the source register files across the banks with FP8 activation data broadcast from the host.

LPDDR5X-PIM Operation : Activation Write (WRPB ¹⁾)

- The following execution process shows how the PIM will conduct a MAC operation (assumption: weight parameters already written into the cell, currently in PIM MB mode), followed by sending calculated data to host

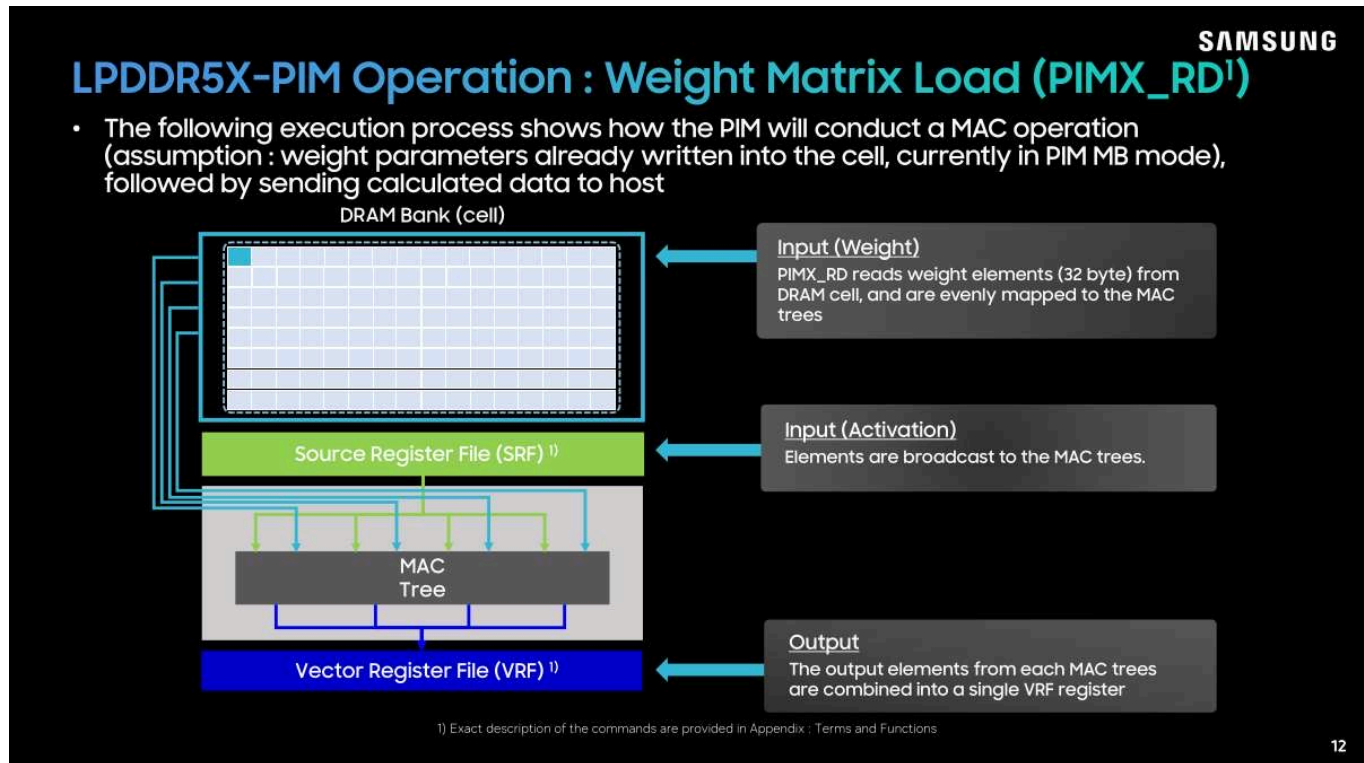


1) Exact description of the commands are provided in Appendix : Terms and Functions
Note: Example shown depicts 512 elements of FP8 data. Considering activation datatype, number of elements will vary (limited to the 4K-bit SRF register size).

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Hot Chips 2026 Samsung LPDDR5X PIM Slide 11 LPDDR5X-PIM Operation: Activation Write (WRPB ¹⁾)

Next, the weight matrix loads. PIMX_RD reads 32-byte weight elements from the DRAM cell, maps them across the MAC trees, and combines the outputs into a single vector register file.

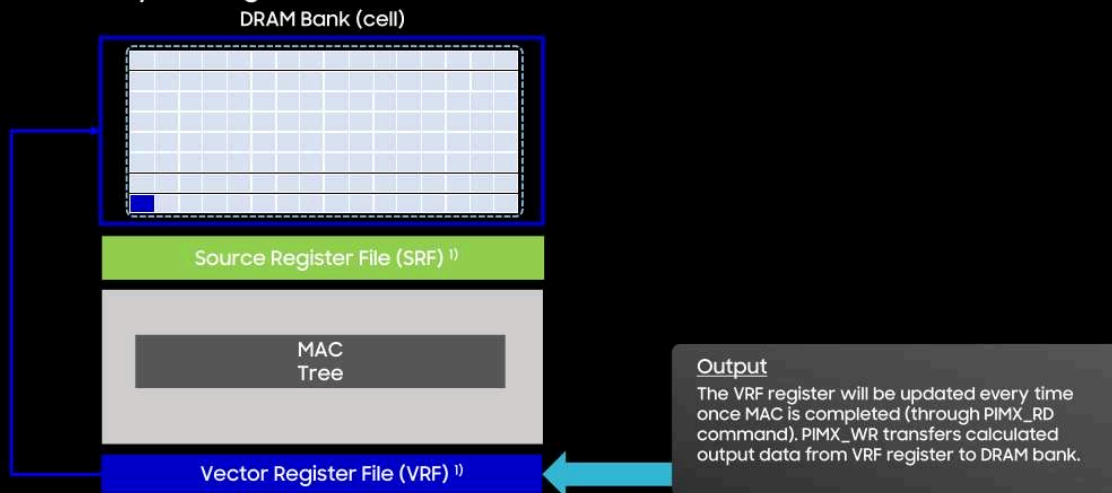


Hot Chips 2026 Samsung LPDDR5X PIM Slide 12 LPDDR5X-PIM Operation: Weight Matrix Load (PIMX_RD1)

Partial sums are written to the vector register file as each MAC completes. PIMX_WR then moves that data back to the DRAM bank, and the read-to-write ratio is free to vary rather than being locked at 1:1.

LPDDR5X-PIM Operation : Vector Store (PIMX_WR¹⁾)

- The following execution process shows how the PIM will conduct a MAC operation (assumption : weight parameters already written into the cell, currently in PIM MB mode), followed by sending calculated data to host



¹⁾ Exact description of the commands are provided in Appendix : Terms and Functions

Note: To move a fully filled VRF register of 1K-bits to the cell, 4 PIMX_WR commands are required. PIMX_RD to PIMX_WR command execution ratio does not have to be 1:1

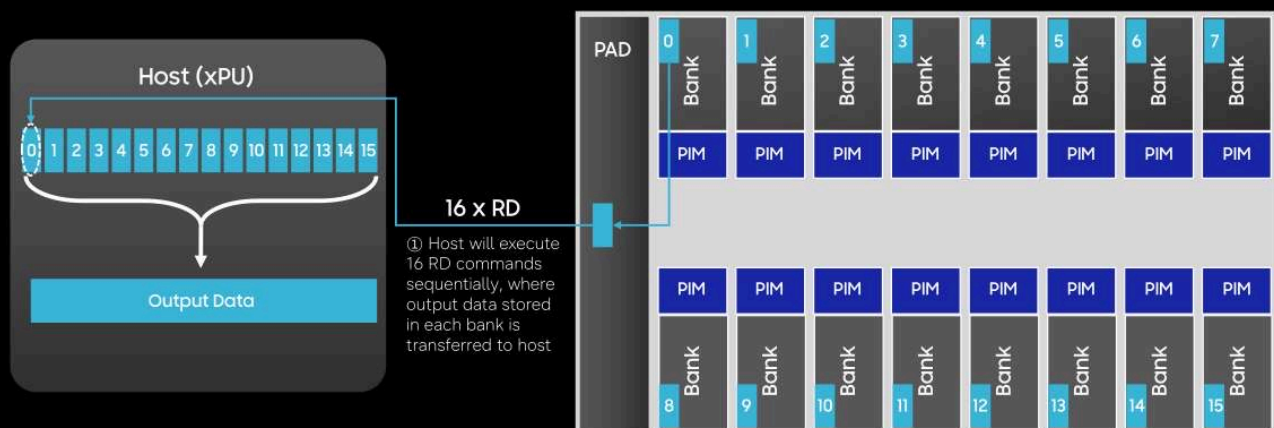
13

Hot Chips 2026 Samsung LPDDR5X PIM Slide 13 LPDDR5X-PIM Operation: Vector Store (PIMX_WR¹)

To pull results back, the host switches to single-bank mode and issues sequential RD commands. A maximum of 64 such reads serves the 1-kbit vector register file.

LPDDR5X-PIM Operation : Vector Read

- From here, the host switches from MB → SB mode
- Here, the host executes 16 RD commands to retrieve calculated PIM data



Note: Example shown depicts 16 RD commands. Based on the size of the VRF register (1K bits), a maximum of 64 RD commands can be executed

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Hot Chips 2026 Samsung LPDDR5X PIM Slide 14 LPDDR5X-PIM Operation: Vector Read

Samsung validated the design on silicon. This evaluation used its edge AI accelerator SoC with both LPDDR5X and LPDDR5X-PIM, running Llama-3.1-8B with a 320-token context, SINT8 activations, SINT4 weights, and SINT32 output.

SAMSUNG

Evaluation Setup

- Tested on silicon w/ Samsung Edge AI accelerator SoC (LPDDR5X & LPDDR5X-PIM)
- Model used for evaluation : Llama-3.1-8B
- Context Length : 320 Tokens (Input & Output)
- Applied data type for Activation / Weight / Output : SINT8 / SINT4 / SINT32

Prompt : *"The challenges of building Processing in Memory (PIM) systems?"*

Samsung Edge AI Accelerator SoC w/ LPDDR5X

Processing in Memory (PIM) systems are designed to accelerate data processing and analysis by performing computations directly on data stored in memory. While PIM systems offer significant performance benefits, they also present several challenges, including:

1. **Memory Hierarchy:** PIM systems need to navigate a complex memory hierarchy, which includes cache, main memory, and storage devices. This requires efficient management of memory accesses and data transfer between different levels of the hierarchy.
2. **Memory Bandwidth:** PIM systems require high-bandwidth memory interfaces to move large amounts of data between memory and processing units. This can be challenging, especially when dealing with large datasets.
3. **Scalability:** As PIM systems are designed to handle massive datasets, scalability becomes a significant challenge. PIM systems need to be able to efficiently process data in parallel, without compromising performance or increasing power consumption.
4. **Energy Efficiency:** PIM systems are designed to be energy-efficient, as they operate at the memory level, which is a power-hungry component. PIM systems need to minimize power consumption while maintaining performance.
5. **Memory Error Handling:** PIM systems need to handle memory errors, such as errors caused by faulty memory or data corruption, to ensure reliable operation.
6. **Data Movement:** PIM systems need to efficiently move data between different levels of the memory hierarchy, which can be challenging, especially.

Time	Output TPS	Tokens
0.51 s	27.0	272

Samsung Edge AI Accelerator SoC w/ LPDDR5X-PIM

Llama v3.1-8B Instruct @ 4096 weight

The challenges of building PIM(Processing in Memory) systems?

Building Processing in Memory (PIM) systems poses several challenges, including:

1. **Scalability:** PIM systems require a scalable architecture and data management capabilities to handle large volumes of data.
2. **Performance:** PIM systems must handle data processing and storage capacity, memory, and speed. The performance of these systems depends on the processing capacity of the memory.
3. **Data Management:** Managing PIM systems requires a proper data management strategy to ensure the data storage is efficient and the PIM system can efficiently process large amounts of data, and also to ensure store large amounts of data.
4. **Data Ingestion:** PIM systems should provide data ingestion and processing and also provide integration to other PIM systems.
5. **Data processing speed:** PIM systems must be fast and processing data storage in a few seconds.

Time	Output TPS	Tokens
0.53 s	81.3	184

*All third-party brand names, product names, and trademarks referenced herein are the property of their respective owners. Such references are solely for the purpose of describing compatibility and does not imply any affiliation with, endorsement by, or sponsorship by the respective trademark or copyright holders.

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Hot Chips 2026 Samsung LPDDR5X PIM Slide 15 Evaluation Setup

This LPDDR5X-PIM build finished the run in 5.4 seconds against 12.3 seconds on conventional LPDDR5X, a 2.28x gain, and output climbed to 81.3 tokens per second from 27.0, a 3.01x gain. That is a big part of the value proposition.

Evaluation Results

- Samsung edge AI accelerator SoC performance comparison (LPDDR5X-PIM, LPDDR5X)



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Hot Chips 2026 Samsung LPDDR5X PIM Slide 16 Evaluation Results

Samsung made clear that software is part of the package. A simulator and datasheet are available on request, and the SDK bundles reference tooling, while LPDDR6-PIM moves toward a finalized JEDEC LP6-PIM specification.

Current Progress in LP-PIM

LPDDR5X-PIM

- Simulator, and datasheet are available upon request
- The simulator is part of our Software Development Kit (SDK) bundle, which can be referred in the appendix
 - SDK is for reference guidance only, as system operation will vary by host
- More information regarding the simulator can be referred to the following paper: "S. Cha, J. Choi, B. Kim, Y. Paik, S. Lee, and K. Sohn, "LP5X-PIM Sim: A High-Fidelity HW/SW Integrated Simulator for LPDDR5X-PIM," [arXiv:2606.00636](https://arxiv.org/abs/2606.00636), May 2026"

LPDDR6-PIM

- JEDEC 1st full version spec near finalization (specific for mobile/client profile)
- Next JEDEC version is also being considered

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Hot Chips 2026 Samsung LPDDR5X PIM Slide 17 Current Progress in LP-PIM

The JEDEC step seems like a big step on the way from a research project to a shipping product.

Final Words

Samsung is positioning LPDDR5X-PIM as a lower-cost, lower-power alternative to HBM for AI inference, and the measured gains on its edge AI accelerator make that pitch concrete. This one feels like one where a large customer is going to have to come in and say this is needed, implement it, and then it will get widely adopted. Perhaps the challenge is that if you give a memory vendor not just control over the memory but also the compute side, they have more leverage with their customers. You need standardization at least so a vendor can second source.